

Personal Statement

For over a decade, I have navigated the dynamic intersection of software engineering, artificial intelligence, and computer vision, building a career defined by turning technical innovation into tangible solutions. Today, I stand at a pivotal crossroads, driven by a profound desire to pivot my expertise toward biomedical engineering (BME)—a field where technological precision can directly impact human health, and where my background as an algorithm engineer can address unmet needs in biological research and clinical practice. Applying for a 2026 PhD position in BME at National University of Singapore (NUS) is not just a career transition; it is the culmination of years of technical growth, deliberate learning, and a deep-seated belief in the transformative power of interdisciplinary collaboration.

My professional journey has been shaped by a relentless focus on translating complex algorithms into practical, high-performance systems. As a software algorithm engineer with over 10 years of experience, I have led end-to-end development across diverse high-stakes projects, each honing skills that I now seek to apply to BME. At Shenzhen SmartMore Information Technology Co., Ltd, I led an AI-driven robotic arm project that integrated Transformer models for action prediction and real-time environmental adaptation—an endeavor that required seamless integration of computer vision, large-model deployment, and hardware-software interaction. This experience taught me to design adaptive systems that bridge theoretical AI with real-world variability, a skill directly transferable to biological systems where dynamic environments and complex data demand flexible technical solutions. Similarly, my work developing a high-precision OCR tool for the semiconductor industry—achieving performance matching or exceeding industry leader Cognex, with 100% recognition rate for fixed formats—showcased my ability to optimize algorithms for extreme precision and reliability, critical qualities in biomedical applications where accuracy can mean the difference between actionable insights and

misleading results.

My foray into biomedical-related technology further solidified my passion for this transition. During a collaboration with BGI Genomics, I designed AI-driven algorithms for real-time inspection of reagent bottle caps and OCR-based character detection on medication bottles, ensuring manufacturing quality and traceability in a life sciences context. This project was a revelation: it showed me how my expertise in defect detection, image analysis, and real-time system integration could directly support biomedical workflows. Later, while pursuing my Master's in Microelectronics Technology & Materials at The Hong Kong Polytechnic University, I delved into molecular dynamics (MD) simulations, gaining hands-on experience with tools like LAMMPS and AI-driven MD platforms through my course project on CsPbBr₃ phase transitions. This academic exposure opened my eyes to the vast potential of merging computational engineering with biological research—how algorithms can unravel molecular mechanisms, simulate cellular processes, and accelerate discoveries in fields like neurobiology and cell engineering.

Recognizing the gap between my engineering background and the biological knowledge required for BME, I have taken proactive steps to build a foundational understanding of the field. I am currently self-studying Prof. Rongwu Yang's *Molecular Biology* course, with plans to follow with Prof. Zhonghe Zhai's *Cell Biology* course. This self-directed learning has not only equipped me with core biological concepts—from neuronal structures to molecular signaling—but has also sharpened my ability to identify where my technical skills can add value: automating the analysis of cellular imaging data, optimizing MD simulation tools for specific biological scenarios, or integrating AI with experimental workflows to reduce human error and accelerate research. My technical toolkit, honed over years of practice, includes proficiency in C/C++/Python programming, AI model deployment (PyTorch, ONNX Runtime), image processing (OpenCV, Halcon), and hardware integration (PCB design, 3D printing, Bluetooth microcontrollers)—a versatile skill set that

allows me to bridge the gap between software, hardware, and biology.

What excites me most about BME is its potential to solve problems that matter. I am particularly drawn to two research directions: Biomedical Imaging, Instrumentation, Sensing and AI, and Molecular, Cellular and Tissue Engineering. In imaging and AI, my 10+ years of experience in image segmentation, target tracking, and automated data analysis can revolutionize how biomedical images are processed—turning raw data from fMRI, cell imaging, or microfluidic systems into actionable insights. For example, building on my work in defect detection and character recognition, I aim to develop AI-driven tools that automate the segmentation of neural structures in imaging data, quantify cellular dynamics, or classify pathological changes with high precision. In molecular and cellular engineering, my experience with MD simulations and system optimization can support efforts to model molecular interactions, simulate cellular processes, and optimize computational tools like GROMACS for biological research—accelerating the understanding of mechanisms such as neurotransmitter release or cellular signaling.

Beyond technical skills, my career has taught me the importance of collaboration and adaptability—critical traits in interdisciplinary research. As a co-founder and CTO of Shenzhen PHDI Co., Ltd, I led the development of a car-mounted Bluetooth controller, overseeing everything from schematic design and PCB layout to 3D printing and software integration. This experience required close collaboration with hardware teams, product managers, and end-users, teaching me to communicate complex technical concepts across disciplines and align solutions with real-world needs. Similarly, in projects with Sony and Apple, I worked with optical teams, client stakeholders, and software engineers to deliver tailored solutions—skills I will bring to collaborations with biologists, clinicians, and fellow BME researchers at NUS. My proficiency in Cantonese, Mandarin, English, and basic Japanese further enables me to bridge cultural and linguistic gaps in cross-team projects, fostering effective teamwork in a global research environment.

My decision to pursue a PhD at NUS is rooted in the university's outstanding reputation in BME and its alignment with my research interests. The opportunity to work with world-class faculty, access state-of-the-art facilities (such as HPC clusters for MD simulations and imaging labs), and collaborate with clinical partners in the Greater Bay Area is invaluable. I am eager to contribute to research that translates technical innovation into clinical impact—whether by developing intelligent imaging tools to support disease diagnosis, optimizing computational models to unravel biological mechanisms, or designing hardware-software integrated systems for personalized medicine.

Looking ahead, my goal as a PhD student is to become a bridge between engineering and biology—someone who can not only develop cutting-edge algorithms but also understand the biological context in which they operate, ensuring that technical solutions address real research and clinical needs. I am committed to embracing the challenges of interdisciplinary learning, continuing to deepen my biological knowledge while leveraging my engineering expertise to drive innovation. Ultimately, I aspire to contribute to advancements that improve human health—whether through more accurate diagnostic tools, more efficient research workflows, or novel therapeutic technologies.

I am confident that my unique combination of technical experience, proactive learning, and passion for BME makes me a strong candidate for your PhD program. I bring not just a skill set, but a track record of delivering results, a willingness to learn, and a vision for how engineering can transform biomedical research. I am eager to join the vibrant research community at NUS, collaborate with leading experts in the field, and dedicate myself to impactful work that bridges technology and biology.

Thank you for considering my application. I look forward to the opportunity to discuss how my background and aspirations align with your lab's research goals.